

Chapter 1

4

Define risk factor

Risk factors is an exposure to something that is associated with a disease, morbidity, mortality, or other adverse health outcome.

What are some examples of risk factors that you experience in your life?

One of the big things in my life is I eat a lot of foods high in sugar and highly processed, which is a risk factor for things like obesity and diabetes. To mitigate this risk factor I try to exercise frequently.

Another risk factor in my life is I participate in Judo club where we practice a lot of Judo and Jujitsu. These sports are close contact and we practice on old wrestling mats. There is a wide variety of risk factors associated with this sport because of a lot of close contact with other people (so if someone is sick many people can get sick) and the mats could be dirty as well allowing germs to spread when rolling on them. To mitigate these risk factors, the club asks that people do not roll if they are sick, we also clean the mats before and after using them, and everyone is strongly advised to thoroughly shower after the club meeting is over.

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What are four uses of epidemiology? (AND RANK THEM)

On the slides the listed uses of epidemiology are as follows: Historical, Community Health, Health Services, Risk Assessment, and Disease Causality.

1. Disease Causality: Determining the causes of diseases by analyzing the experiences and environment of the infected population. An example of this is John Snow using disease causality to determine that cholera was caused by the infected water well.
2. Health Services: Studying the current state of health services and ways to improve their effectiveness. The example given in the slides is optimizing the placement of health services to increase the total coverage of areas but another metric that could be studied is the average duration patients are in the hospital to determine the efficiency of an individual health service area.
3. Risk Assessment: Determining an individual's risk of disease / accident / negative health outcome and trying to prevent / mitigate these outcomes. An example of this is at a yearly doctor checkup they know your medical history and the medical history of your family and if say diabetes runs in the family they can check you for diabetes while also offer ways to prevent you from getting diabetes.

4. Community Health: The analysis of current overall health of the community using a variety of census statistics. An example of this could be a report of flu cases near college campuses spiking during the winter then suddenly dropping during winter break (because students left campus area to go home).

Why did you rank them in that order?

I ordered them based on my opinion on 2 factors: impact scale (individuals vs communities vs global) and use type (analytical, preventative, mitigative). For ordering, I think that the impact scale is ordered by more people means more important, and for use type my order is 1. preventative, 2. mitigative, 3. analytical because I believe that preventing a disease from spreading is more important than mitigating the effects of a disease. In my opinion the disease causality use is the number one because its impact scale is number 1 and its use type is both preventative and mitigative. Health Services is both preventative and mitigative as well but mainly looks at communities. Risk assessment is at an individual scale and is preventative and mitigative use types. Lastly community health is last because it is community level and is mostly analytical which in my opinion is the least important use type.

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Why do people regard John Snow as the Father of epidemiology? Provide 3 reasons

John Snow demonstrated the powers of observation and written expression.

John Snow applied a variety of epidemiologic methods such as mapping cholera infections on top of a map of water suppliers to determine that the one supplier had higher infections than the other

Recommended a public health measures to prevent disease

Chapter 2

2

What is Quantitative Data?

Quantitative data is numerable / orderable data that can be represented with numbers and strictly ordered.

What is Qualitative Data?

Qualitative data is categorical data where the data is put into different categories.

Categorize the following parameters:

- Sex

Qualitative

- Race

Qualitative

- Weight

Quantitative

4

Define the following and cite their applications

1. Stratified random sampling

Stratified random sampling is creating strata of individual subgroups of the population and ensuring random sampling of each strata. An application of this could be to ensure each strata is sampled relatively equally in order to make accurate comparisons across the strata. For example, for facial recognition software, it is ideal to use stratified random sampling to ensure each strata (in this case race) is thoroughly represented in the dataset to ensure that the functionality of the software is not dependent on race.

2. Systemic sampling

Systemic sampling is choosing data samples from a predefined system such as using every other data point. The primary issue with systemic sampling is under representation of the data set by not sampling later samples. Applications of this could include trying to obtain a specific amount of data points.

3. Convenience sampling

Convenience sampling revolves around sampling data as convenient. What this can look like is asking friends and those convenient to ask, typically resulting in a biased sample set. The primary application is to quickly collect data and use that to determine if additional study / research should be conducted.

4. Cluster sampling

Cluster sampling is sampling from a single cluster. The example given in the book is sampling from a single city block. This sampling method is intentionally biased towards the clustered population and should not be used as an overarching representation of the population. An example of cluster sampling is John Snow's cholera map on the slides which samples only a few blocks of people and used this to create a small map of infections vs the water wells.

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Why is random sampling unbiased?

Samples are selected via a random process thus are unbiased because the samples aren't clustered around any particular group.

Chapter 3

7

How are incidence and prevalence related to each other?

Incidence refers to the number of new incidents of a given disease within a population at a certain time while prevalence is the number of people that are ill with a particular disease within a population at a certain time. In other words, incidence refers to new cases while prevalence is the total cases. As such, incidence will increase prevalence but prevalence may not necessarily increase incidence. For example, if an entire population is infected the incidence rate will not be high because there are no new cases while the prevalence will be super high.

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Disease	2016	2017	2018	2019
Butilsm	31	19	17	20
Cholera	15	10	14	14
Leprosy	78	94	90	77
Rabies	0	2	3	0

What quantitative measure is shown in the table (3.3)?

Table 3.3 has the number of cases of certain diseases (Botulism, Cholera, Hansen's Disease, Rabies) in the US over the years 2016-2019.

Describe the annual trends in the case of infrequently reported notifiable diseases

For Botulism, there was a sharp drop of 33% from 2016 (31 cases) to 2017 (19 cases) and remained relatively steady from 2017 to 2019 (20 cases)

Cholera: the number of Cholera cases has remained relatively consistent at 14 cases per year except for 2017 which had 10 cases.

Leprosy started at 78 cases in 2016, then increased by roughly 20% in 2017 to 94 cases then 2018 there were similar cases amount of cases as 2017 but then in 2019 there was a decrease in cases back to 77.

Rabies had 0 cases in 2016 and increased to 2/3 for 2017 and 2018 but then decreased back to 0 in 2019

Why do we have to be careful about comparing counts across different populations?

Different populations have different levels of normal expectancy of certain diseases. For example, a certain population might have a malaria Endemic (high rates of malaria is normal for the population) while other populations might not have a malaria Endemic. This could be due to a wide variety of factors such as presence of mosquitos within the population.

Problem

Calculate the Incidence Rate

$$\frac{\# \text{ of incidents}}{\# \text{ of ppl}} \times 100,000 = \frac{441}{11.42 \times 10^6} \times 100,000 = 3.8616 \text{ per } 100,000$$

Calculate the Fatality Rate

$$\frac{\# \text{ of deaths}}{\# \text{ of cases}} = \frac{31}{441} * 100 = 7.029$$